

NKPC HSA Bayesian State-Space Estimation: Inverse markup gap

June 22, 2026

1 Data and Transformations

Raw data are read from `data/raw/`. Processed model-ready data are written to `data/processed/`. The HSA competition aggregator convention is $N_t = 100 \log(N_t^{level})$ unless a run metadata file explicitly states that the series was already transformed.

2 Model Specifications

The CES benchmark is

$$\pi_t = \alpha\pi_{t-1} + (1 - \alpha)E_t\pi_{t+1} + \kappa x_t + e_t.$$

The dynamic HSA specification is

$$\pi_t = \alpha\pi_{t-1} + (1 - \alpha)E_t\pi_{t+1} + \kappa x_t - \theta\hat{N}_t + e_t.$$

The steady HSA specification is

$$\pi_t = \alpha\pi_{t-1} + (1 - \alpha)E_t\pi_{t+1} + (\kappa_0 + \delta\bar{N}_t)x_t + e_t.$$

The both-effects HSA specification is

$$\pi_t = \alpha\pi_{t-1} + (1 - \alpha)E_t\pi_{t+1} + (\kappa_0 + \delta\bar{N}_t)x_t - (\theta_0 + \gamma\bar{N}_t)\hat{N}_t + e_t.$$

The maintained HSA decomposition is

$$N_t = \hat{N}_t + \bar{N}_t, \quad \hat{N}_t = \rho_1\hat{N}_{t-1} + \rho_2\hat{N}_{t-2} + u_t, \quad \bar{N}_t = n + \bar{N}_{t-1} + \epsilon_t.$$

3 Unit and Covariance Conventions

Priors for κ , κ_0 , and δ are specified in physical units. Legacy Gibbs samplers internally use `KAPPA_SCALE=100` when the design matrix contains $x_t/100$. All posterior draws, tables, figures, SDDR calculations, and model-comparison summaries are reported in physical units. The default dynamic-HSA shock covariance allows only (e_t, ζ_t) correlation; diagonal and historical full covariance modes are available only through explicit run metadata.

4 Prior Specifications

Baseline, weak, and tight priors are stored in `configs/`. Robustness tables below summarize whether conclusions for δ , θ , and γ change across these choices.

5 1. Posterior Mean Coefficients

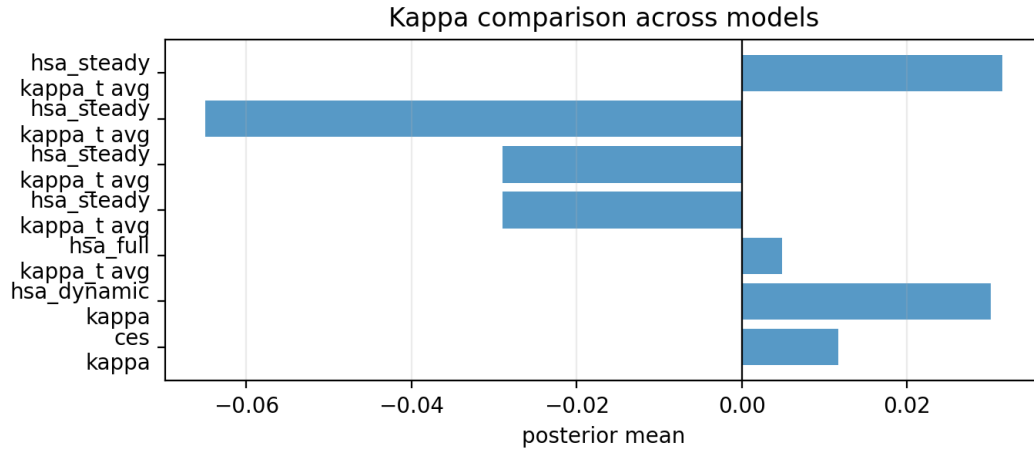
This section reports posterior means and credible intervals for the main scalar coefficients in each estimated run.

run	model	parameter	posterior_mean	posterior_sd	ci.2.5	ci.97.5
ces_inv_markup_baseline.20260622.131545	ces	alpha	0.6342	0.0588	0.5474	0.7428
ces_inv_markup_baseline.20260622.131545	ces	kappa	0.0116	0.1238	-0.2051	0.3018
ces_inv_markup_baseline.20260622.131545	ces	phi.1	0.6733	0.0721	0.5321	0.7739
ces_inv_markup_baseline.20260622.131545	ces	lambda.ez	-0.0370	0.1484	-0.3385	0.2248
hsa_dynamic_inv_markup_baseline.20260622.131549	hsa_dynamic	alpha	0.6161	0.0515	0.5294	0.7235
hsa_dynamic_inv_markup_baseline.20260622.131549	hsa_dynamic	kappa	0.0301	0.1531	-0.2127	0.3024
hsa_dynamic_inv_markup_baseline.20260622.131549	hsa_dynamic	theta	0.0083	0.0039	0.0029	0.0158
hsa_dynamic_inv_markup_baseline.20260622.131549	hsa_dynamic	phi.1	0.6507	0.0710	0.5172	0.7831
hsa_dynamic_inv_markup_baseline.20260622.131549	hsa_dynamic	rho.1	1.8713	0.0303	1.8074	1.9099
hsa_dynamic_inv_markup_baseline.20260622.131549	hsa_dynamic	rho.2	-0.8748	0.0301	-0.9150	-0.8101
hsa_dynamic_inv_markup_baseline.20260622.131549	hsa_dynamic	n	-0.0353	0.0486	-0.1143	0.0509
hsa_full_inv_markup_baseline.20260622.131551	hsa_full	alpha	0.6169	0.0546	0.5257	0.7203
hsa_full_inv_markup_baseline.20260622.131551	hsa_full	kappa.0	0.0961	0.1934	-0.2090	0.4047
hsa_full_inv_markup_baseline.20260622.131551	hsa_full	delta	-0.0004	0.0012	-0.0028	0.0013
hsa_full_inv_markup_baseline.20260622.131551	hsa_full	theta.0	0.0669	0.2020	-0.2423	0.3712
hsa_full_inv_markup_baseline.20260622.131551	hsa_full	gamma	-0.0001	0.0010	-0.0015	0.0015
hsa_full_inv_markup_baseline.20260622.131551	hsa_full	phi.1	0.6639	0.0583	0.5845	0.7842
hsa_full_inv_markup_baseline.20260622.131551	hsa_full	rho.1	1.1914	0.1130	1.0022	1.3879
hsa_full_inv_markup_baseline.20260622.131551	hsa_full	rho.2	-0.2695	0.1199	-0.4699	-0.0663
hsa_full_inv_markup_baseline.20260622.131551	hsa_full	n	-0.0899	0.0590	-0.1805	0.0371
hsa_full_inv_markup_baseline.20260622.131551	hsa_full	lambda.ez	-0.0122	0.1928	-0.3014	0.2785
hsa_steady_inv_markup_baseline.20260622.131547	hsa_steady	alpha	0.6484	0.0637	0.5539	0.7759
hsa_steady_inv_markup_baseline.20260622.131547	hsa_steady	kappa.0	0.1008	0.1901	-0.1878	0.4623
hsa_steady_inv_markup_baseline.20260622.131547	hsa_steady	delta	-0.0006	0.0012	-0.0028	0.0021
hsa_steady_inv_markup_baseline.20260622.131547	hsa_steady	phi.1	0.6665	0.0599	0.5612	0.7780
hsa_steady_inv_markup_baseline.20260622.131547	hsa_steady	rho.1	1.8054	0.0517	1.7062	1.8953
hsa_steady_inv_markup_baseline.20260622.131547	hsa_steady	rho.2	-0.8090	0.0517	-0.8973	-0.7095
hsa_steady_inv_markup_baseline.20260622.131547	hsa_steady	n	-0.0538	0.0713	-0.1901	0.0504
hsa_steady_inv_markup_baseline.20260622.131547	hsa_steady	lambda.ez	0.0197	0.1538	-0.1838	0.3257
hsa_steady_inv_markup_baseline.20260622.132032	hsa_steady	alpha	0.6484	0.0637	0.5539	0.7759
hsa_steady_inv_markup_baseline.20260622.132032	hsa_steady	kappa.0	0.1008	0.1901	-0.1878	0.4623
hsa_steady_inv_markup_baseline.20260622.132032	hsa_steady	delta	-0.0006	0.0012	-0.0028	0.0021
hsa_steady_inv_markup_baseline.20260622.132032	hsa_steady	phi.1	0.6665	0.0599	0.5612	0.7780
hsa_steady_inv_markup_baseline.20260622.132032	hsa_steady	rho.1	1.8054	0.0517	1.7062	1.8953
hsa_steady_inv_markup_baseline.20260622.132032	hsa_steady	rho.2	-0.8090	0.0517	-0.8973	-0.7095
hsa_steady_inv_markup_baseline.20260622.132032	hsa_steady	n	-0.0538	0.0713	-0.1901	0.0504
hsa_steady_inv_markup_baseline.20260622.132032	hsa_steady	lambda.ez	0.0197	0.1538	-0.1838	0.3257
hsa_steady_inv_markup_tight.20260622.132037	hsa_steady	alpha	0.6124	0.0568	0.5147	0.7208
hsa_steady_inv_markup_tight.20260622.132037	hsa_steady	kappa.0	0.0996	0.1082	-0.1344	0.2868
hsa_steady_inv_markup_tight.20260622.132037	hsa_steady	delta	-0.0008	0.0008	-0.0023	0.0005
hsa_steady_inv_markup_tight.20260622.132037	hsa_steady	phi.1	0.6742	0.0544	0.5698	0.7500
hsa_steady_inv_markup_tight.20260622.132037	hsa_steady	rho.1	0.5521	0.0804	0.3924	0.6863
hsa_steady_inv_markup_tight.20260622.132037	hsa_steady	rho.2	-0.3597	0.1083	-0.5528	-0.1244
hsa_steady_inv_markup_tight.20260622.132037	hsa_steady	n	-0.0193	0.0447	-0.0985	0.0567
hsa_steady_inv_markup_tight.20260622.132037	hsa_steady	lambda.ez	0.0315	0.1392	-0.2849	0.2151
hsa_steady_inv_markup_weak.20260622.132035	hsa_steady	alpha	0.6458	0.0559	0.5432	0.7534
hsa_steady_inv_markup_weak.20260622.132035	hsa_steady	kappa.0	0.2063	0.5646	-0.7736	1.3392
hsa_steady_inv_markup_weak.20260622.132035	hsa_steady	delta	-0.0009	0.0029	-0.0066	0.0041
hsa_steady_inv_markup_weak.20260622.132035	hsa_steady	phi.1	0.6587	0.0661	0.5477	0.7849
hsa_steady_inv_markup_weak.20260622.132035	hsa_steady	rho.1	1.9069	0.0397	1.8328	1.9713
hsa_steady_inv_markup_weak.20260622.132035	hsa_steady	rho.2	-0.9096	0.0394	-0.9730	-0.8367
hsa_steady_inv_markup_weak.20260622.132035	hsa_steady	n	-0.1962	0.0958	-0.3445	-0.0156
hsa_steady_inv_markup_weak.20260622.132035	hsa_steady	lambda.ez	-0.0349	0.2343	-0.5064	0.3567

6 2. Kappa Comparison Across Models

For CES models this is κ . For steady or full HSA models the table also reports κ_0 , δ , and the average path of $\kappa_t = \kappa_0 + \delta \bar{N}_t$ when available.

run	model	kappa_mean	kappa_ci.2.5	kappa_ci.97.5	kappa_0_mean	kappa_0_ci.2.5	kappa_0_ci.97.5	delta_mean	delta_ci.2.5	delta_ci.97.5	kappa_t_0verall_mean	kappa_t_start_mean	kappa_t_end_mean
ces_inv_markup_baseline.20260622.131545	ces	0.0116	-0.2051	0.3018	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
hsa_dynamic_inv_markup_baseline.20260622.131549	hsa_dynamic	0.0301	-0.2127	0.3024	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
hsa_full_inv_markup_baseline.20260622.131551	hsa_full	NaN	NaN	0.0961	-0.2090	0.4047	-0.0004	-0.0028	0.0013	0.0049	0.0066	0.0163	0.0163
hsa_steady_inv_markup_baseline.20260622.131547	hsa_steady	NaN	NaN	NaN	-0.1878	0.4623	-0.0006	-0.0028	0.0021	-0.0280	-0.0284	-0.0286	-0.0286
hsa_steady_inv_markup_tight.20260622.132037	hsa_steady	NaN	NaN	NaN	-0.1878	0.4623	-0.0006	-0.0028	0.0021	-0.0280	-0.0284	-0.0286	-0.0286
hsa_steady_inv_markup_weak.20260622.132035	hsa_steady	NaN	NaN	NaN	-0.1344	0.2868	-0.0008	-0.0023	0.0005	-0.0649	-0.0628	-0.0433	-0.0433
hsa_steady_inv_markup_weak.20260622.132035	hsa_steady	NaN	NaN	NaN	0.2063	-0.7736	1.3392	-0.0009	-0.0066	0.0041	0.0315	0.0217	0.0411

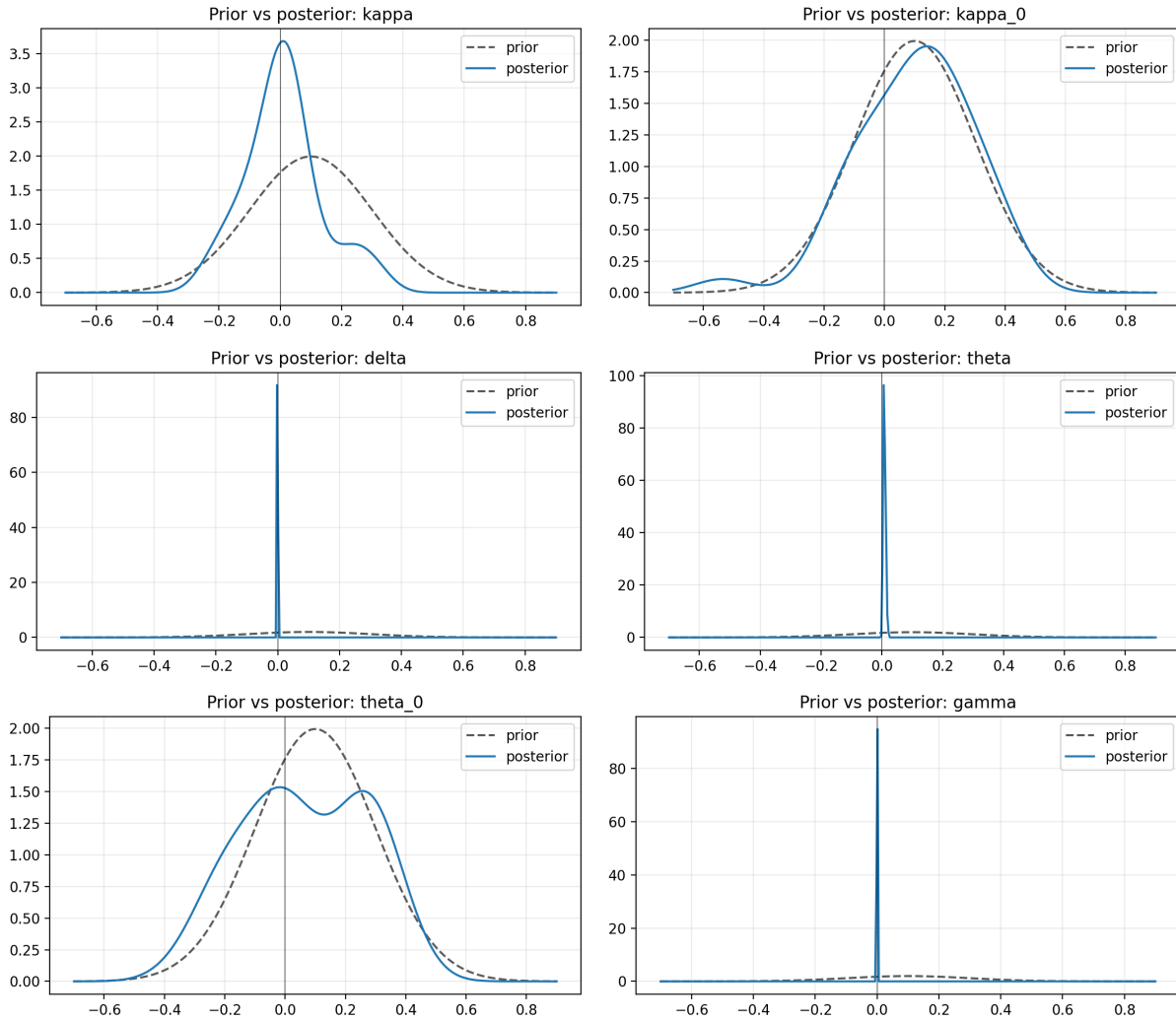


7 3. Time-Varying Coefficient Paths

The main time-varying coefficients are κ_t and, in the both-effects model, $\theta_t = \theta_0 + \gamma \bar{N}_t$.

run	model	coefficient	time_index	posterior_mean	ci.2.5	ci.97.5
hsa_full_inv_markup_baseline.20260622.131551	hsa_full	kappa.t	0	0.0066	-0.2587	0.2588
hsa_full_inv_markup_baseline.20260622.131551	hsa_full	kappa.t	5	0.0063	-0.2607	0.2581
hsa_full_inv_markup_baseline.20260622.131551	hsa_full	kappa.t	11	0.0042	-0.2632	0.2631
hsa_full_inv_markup_baseline.20260622.131551	hsa_full	kappa.t	16	-0.0003	-0.2704	0.2701
hsa_full_inv_markup_baseline.20260622.131551	hsa_full	kappa.t	21	-0.0016	-0.2742	0.2724
hsa_full_inv_markup_baseline.20260622.131551	hsa_full	kappa.t	27	-0.0013	-0.2730	0.2717
hsa_full_inv_markup_baseline.20260622.131551	hsa_full	kappa.t	32	-0.0008	-0.2729	0.2708
hsa_full_inv_markup_baseline.20260622.131551	hsa_full	kappa.t	37	-0.0004	-0.2714	0.2703
hsa_full_inv_markup_baseline.20260622.131551	hsa_full	kappa.t	43	-0.0016	-0.2733	0.2740
hsa_full_inv_markup_baseline.20260622.131551	hsa_full	kappa.t	48	-0.0028	-0.2761	0.2774
hsa_full_inv_markup_baseline.20260622.131551	hsa_full	kappa.t	53	-0.0038	-0.2785	0.2769
hsa_full_inv_markup_baseline.20260622.131551	hsa_full	kappa.t	59	-0.0038	-0.2788	0.2775
hsa_full_inv_markup_baseline.20260622.131551	hsa_full	kappa.t	64	-0.0018	-0.2759	0.2732
hsa_full_inv_markup_baseline.20260622.131551	hsa_full	kappa.t	70	0.0023	-0.2680	0.2661
hsa_full_inv_markup_baseline.20260622.131551	hsa_full	kappa.t	75	0.0056	-0.2621	0.2616
hsa_full_inv_markup_baseline.20260622.131551	hsa_full	kappa.t	80	0.0087	-0.2555	0.2559
hsa_full_inv_markup_baseline.20260622.131551	hsa_full	kappa.t	86	0.0103	-0.2526	0.2529
hsa_full_inv_markup_baseline.20260622.131551	hsa_full	kappa.t	91	0.0108	-0.2505	0.2508
hsa_full_inv_markup_baseline.20260622.131551	hsa_full	kappa.t	96	0.0113	-0.2501	0.2504
hsa_full_inv_markup_baseline.20260622.131551	hsa_full	kappa.t	102	0.0130	-0.2486	0.2477
hsa_full_inv_markup_baseline.20260622.131551	hsa_full	kappa.t	107	0.0147	-0.2455	0.2460
hsa_full_inv_markup_baseline.20260622.131551	hsa_full	kappa.t	112	0.0151	-0.2446	0.2458
hsa_full_inv_markup_baseline.20260622.131551	hsa_full	kappa.t	118	0.0162	-0.2427	0.2447
hsa_full_inv_markup_baseline.20260622.131551	hsa_full	kappa.t	123	0.0163	-0.2420	0.2447
hsa_full_inv_markup_baseline.20260622.131551	hsa_full	theta.t	0	0.0465	0.0089	0.0994
hsa_full_inv_markup_baseline.20260622.131551	hsa_full	theta.t	5	0.0464	0.0094	0.0994
hsa_full_inv_markup_baseline.20260622.131551	hsa_full	theta.t	11	0.0455	0.0082	0.1035
hsa_full_inv_markup_baseline.20260622.131551	hsa_full	theta.t	16	0.0447	-0.0036	0.1081
hsa_full_inv_markup_baseline.20260622.131551	hsa_full	theta.t	21	0.0447	-0.0067	0.1102
hsa_full_inv_markup_baseline.20260622.131551	hsa_full	theta.t	27	0.0447	-0.0061	0.1095
hsa_full_inv_markup_baseline.20260622.131551	hsa_full	theta.t	32	0.0447	-0.0039	0.1094
hsa_full_inv_markup_baseline.20260622.131551	hsa_full	theta.t	37	0.0447	-0.0038	0.1092
hsa_full_inv_markup_baseline.20260622.131551	hsa_full	theta.t	43	0.0444	-0.0071	0.1109
hsa_full_inv_markup_baseline.20260622.131551	hsa_full	theta.t	48	0.0440	-0.0122	0.1120
hsa_full_inv_markup_baseline.20260622.131551	hsa_full	theta.t	53	0.0442	-0.0132	0.1137
hsa_full_inv_markup_baseline.20260622.131551	hsa_full	theta.t	59	0.0442	-0.0166	0.1123
hsa_full_inv_markup_baseline.20260622.131551	hsa_full	theta.t	64	0.0446	-0.0070	0.1101
hsa_full_inv_markup_baseline.20260622.131551	hsa_full	theta.t	70	0.0456	-0.0006	0.1045
hsa_full_inv_markup_baseline.20260622.131551	hsa_full	theta.t	75	0.0463	0.0109	0.0995
hsa_full_inv_markup_baseline.20260622.131551	hsa_full	theta.t	80	0.0469	0.0035	0.0953
hsa_full_inv_markup_baseline.20260622.131551	hsa_full	theta.t	86	0.0470	-0.0014	0.0970
hsa_full_inv_markup_baseline.20260622.131551	hsa_full	theta.t	91	0.0471	-0.0036	0.0965
hsa_full_inv_markup_baseline.20260622.131551	hsa_full	theta.t	96	0.0474	-0.0033	0.0986
hsa_full_inv_markup_baseline.20260622.131551	hsa_full	theta.t	102	0.0477	-0.0067	0.1012
hsa_full_inv_markup_baseline.20260622.131551	hsa_full	theta.t	107	0.0484	-0.0095	0.1040
hsa_full_inv_markup_baseline.20260622.131551	hsa_full	theta.t	112	0.0481	-0.0106	0.1050
hsa_full_inv_markup_baseline.20260622.131551	hsa_full	theta.t	118	0.0485	-0.0119	0.1058
hsa_full_inv_markup_baseline.20260622.131551	hsa_full	theta.t	123	0.0487	-0.0123	0.1066
hsa_steady_inv_markup_baseline.20260622.131547	hsa_steady	kappa.t	0	-0.0294	-0.2596	0.2386
hsa_steady_inv_markup_baseline.20260622.131547	hsa_steady	kappa.t	5	-0.0293	-0.2582	0.2387
hsa_steady_inv_markup_baseline.20260622.131547	hsa_steady	kappa.t	11	-0.0293	-0.2567	0.2392
hsa_steady_inv_markup_baseline.20260622.131547	hsa_steady	kappa.t	16	-0.0296	-0.2579	0.2403
hsa_steady_inv_markup_baseline.20260622.131547	hsa_steady	kappa.t	21	-0.0294	-0.2590	0.2405
hsa_steady_inv_markup_baseline.20260622.131547	hsa_steady	kappa.t	27	-0.0292	-0.2611	0.2385
hsa_steady_inv_markup_baseline.20260622.131547	hsa_steady	kappa.t	32	-0.0292	-0.2620	0.2375
hsa_steady_inv_markup_baseline.20260622.131547	hsa_steady	kappa.t	37	-0.0292	-0.2594	0.2361
hsa_steady_inv_markup_baseline.20260622.131547	hsa_steady	kappa.t	43	-0.0291	-0.2614	0.2349
hsa_steady_inv_markup_baseline.20260622.131547	hsa_steady	kappa.t	48	-0.0292	-0.2604	0.2346
hsa_steady_inv_markup_baseline.20260622.131547	hsa_steady	kappa.t	53	-0.0294	-0.2614	0.2348
hsa_steady_inv_markup_baseline.20260622.131547	hsa_steady	kappa.t	59	-0.0297	-0.2617	0.2346
hsa_steady_inv_markup_baseline.20260622.131547	hsa_steady	kappa.t	64	-0.0295	-0.2584	0.2336
hsa_steady_inv_markup_baseline.20260622.131547	hsa_steady	kappa.t	70	-0.0290	-0.2528	0.2344
hsa_steady_inv_markup_baseline.20260622.131547	hsa_steady	kappa.t	75	-0.0289	-0.2510	0.2332
hsa_steady_inv_markup_baseline.20260622.131547	hsa_steady	kappa.t	80	-0.0287	-0.2482	0.2314
hsa_steady_inv_markup_baseline.20260622.131547	hsa_steady	kappa.t	86	-0.0285	-0.2463	0.2326
hsa_steady_inv_markup_baseline.20260622.131547	hsa_steady	kappa.t	91	-0.0285	-0.2459	0.2299
hsa_steady_inv_markup_baseline.20260622.131547	hsa_steady	kappa.t	96	-0.0286	-0.2464	0.2294
hsa_steady_inv_markup_baseline.20260622.131547	hsa_steady	kappa.t	102	-0.0286	-0.2453	0.2274
hsa_steady_inv_markup_baseline.20260622.131547	hsa_steady	kappa.t	107	-0.0285	-0.2446	0.2275
hsa_steady_inv_markup_baseline.20260622.131547	hsa_steady	kappa.t	112	-0.0284	-0.2437	0.2275
hsa_steady_inv_markup_baseline.20260622.131547	hsa_steady	kappa.t	118	-0.0284	-0.2502	0.2275
hsa_steady_inv_markup_baseline.20260622.131547	hsa_steady	kappa.t	123	-0.0286	-0.2462	0.2266
hsa_steady_inv_markup_baseline.20260622.132032	hsa_steady	kappa.t	0	-0.0294	-0.2596	0.2386
hsa_steady_inv_markup_baseline.20260622.132032	hsa_steady	kappa.t	5	-0.0293	-0.2582	0.2387
hsa_steady_inv_markup_baseline.20260622.132032	hsa_steady	kappa.t	11	-0.0293	-0.2567	0.2392
hsa_steady_inv_markup_baseline.20260622.132032	hsa_steady	kappa.t	16	-0.0296	-0.2579	0.2403
hsa_steady_inv_markup_baseline.20260622.132032	hsa_steady	kappa.t	21	-0.0294	-0.2590	0.2405
hsa_steady_inv_markup_baseline.20260622.132032	hsa_steady	kappa.t	27	-0.0292	-0.2611	0.2385

run	model	restriction	prior_mean	prior_sd	sddr_bf01
ces_inv_markup_baseline_20260622_131545	ces	kappa=0	0.1000	0.2000	2.0791
hsa_dynamic_inv_markup_baseline_20260622_131549	hsa_dynamic	kappa=0	0.1000	0.2000	1.3783
hsa_dynamic_inv_markup_baseline_20260622_131549	hsa_dynamic	theta=0	0.1000	0.2000	4.5471
hsa_full_inv_markup_baseline_20260622_131551	hsa_full	kappa_0=0	0.1000	0.2000	0.8905
hsa_full_inv_markup_baseline_20260622_131551	hsa_full	delta=0	0.1000	0.2000	158.5492
hsa_full_inv_markup_baseline_20260622_131551	hsa_full	theta_0=0	0.1000	0.2000	0.8677
hsa_full_inv_markup_baseline_20260622_131551	hsa_full	gamma=0	0.1000	0.2000	180.6376
hsa_steady_inv_markup_baseline_20260622_131547	hsa_steady	kappa_0=0	0.1000	0.2000	1.1450
hsa_steady_inv_markup_baseline_20260622_131547	hsa_steady	delta=0	0.1000	0.2000	144.3115
hsa_steady_inv_markup_baseline_20260622_132032	hsa_steady	kappa_0=0	0.1000	0.2000	1.1450
hsa_steady_inv_markup_baseline_20260622_132032	hsa_steady	delta=0	0.1000	0.2000	144.3115
hsa_steady_inv_markup_tight_20260622_132037	hsa_steady	kappa_0=0	0.1000	0.2000	1.0298
hsa_steady_inv_markup_tight_20260622_132037	hsa_steady	delta=0	0.1000	0.2000	178.8126
hsa_steady_inv_markup_weak_20260622_132035	hsa_steady	kappa_0=0	0.1000	0.2000	0.3672
hsa_steady_inv_markup_weak_20260622_132035	hsa_steady	delta=0	0.1000	0.2000	67.1558



10 Diagnostics and Robustness

run	model	data.spec	prior.spec	parameter	mean	sd	hdi.2.5%	hdi.97.5%	mcmc.mean	ess.bulk	ess.tail	r.hat	warning
ces_inv_markup_baseline.20260622.131545	ces	inv_markup	baseline	alpha	0.6342	0.0588	0.5474	0.7428	NaN	NaN	NaN	NaN	NaN
ces_inv_markup_baseline.20260622.131545	ces	inv_markup	baseline	kappa	0.0116	0.1238	-0.2051	0.3018	NaN	NaN	NaN	NaN	NaN
ces_inv_markup_baseline.20260622.131545	ces	inv_markup	baseline	phi.1	0.6733	0.0721	0.5321	0.7739	NaN	NaN	NaN	NaN	NaN
ces_inv_markup_baseline.20260622.131545	ces	inv_markup	baseline	lambda.ez	-0.0370	0.1484	-0.3385	0.2248	NaN	NaN	NaN	NaN	NaN
ces_inv_markup_baseline.20260622.131545	ces	inv_markup	baseline	rho	-0.0318	0.1111	-0.2636	0.1689	NaN	NaN	NaN	NaN	NaN
ces_inv_markup_baseline.20260622.131545	ces	inv_markup	baseline	sigma.e	0.6637	0.0436	0.5888	0.7183	NaN	NaN	NaN	NaN	NaN
ces_inv_markup_baseline.20260622.131545	ces	inv_markup	baseline	sigma.zeta	0.5032	0.0302	0.4542	0.5548	NaN	NaN	NaN	NaN	NaN
hsa_dynamic_inv_markup_baseline.20260622.131549	hsa_dynamic	inv_markup	baseline	alpha	0.6161	0.0515	0.5294	0.7235	NaN	NaN	NaN	NaN	NaN
hsa_dynamic_inv_markup_baseline.20260622.131549	hsa_dynamic	inv_markup	baseline	kappa	0.0301	0.1531	-0.2127	0.3024	NaN	NaN	NaN	NaN	NaN
hsa_dynamic_inv_markup_baseline.20260622.131549	hsa_dynamic	inv_markup	baseline	theta	0.0083	0.0039	0.0029	0.0158	NaN	NaN	NaN	NaN	NaN
hsa_dynamic_inv_markup_baseline.20260622.131549	hsa_dynamic	inv_markup	baseline	phi.1	0.6507	0.0710	0.5172	0.7831	NaN	NaN	NaN	NaN	NaN
hsa_dynamic_inv_markup_baseline.20260622.131549	hsa_dynamic	inv_markup	baseline	rho.1	1.8713	0.0303	1.8074	1.9099	NaN	NaN	NaN	NaN	NaN
hsa_dynamic_inv_markup_baseline.20260622.131549	hsa_dynamic	inv_markup	baseline	rho.2	-0.8748	0.0301	-0.9150	-0.8101	NaN	NaN	NaN	NaN	NaN
hsa_dynamic_inv_markup_baseline.20260622.131549	hsa_dynamic	inv_markup	baseline	n	-0.0353	0.0486	-0.1143	0.0509	NaN	NaN	NaN	NaN	NaN
hsa_dynamic_inv_markup_baseline.20260622.131549	hsa_dynamic	inv_markup	baseline	sigma.e	0.6541	0.0416	0.5910	0.7451	NaN	NaN	NaN	NaN	NaN
hsa_dynamic_inv_markup_baseline.20260622.131549	hsa_dynamic	inv_markup	baseline	sigma.zeta	0.4868	0.0369	0.4358	0.5681	NaN	NaN	NaN	NaN	NaN
hsa_dynamic_inv_markup_baseline.20260622.131549	hsa_dynamic	inv_markup	baseline	sigma.u	0.4105	0.0376	0.3463	0.4705	NaN	NaN	NaN	NaN	NaN
hsa_dynamic_inv_markup_baseline.20260622.131549	hsa_dynamic	inv_markup	baseline	sigma.eps	0.2340	0.0306	0.1829	0.3093	NaN	NaN	NaN	NaN	NaN
hsa_full_inv_markup_baseline.20260622.131551	hsa_full	inv_markup	baseline	sigma.N	0.2772	0.0289	0.2411	0.3350	NaN	NaN	NaN	NaN	NaN
hsa_full_inv_markup_baseline.20260622.131551	hsa_full	inv_markup	baseline	alpha	0.6169	0.0546	0.5257	0.7203	NaN	NaN	NaN	NaN	NaN
hsa_full_inv_markup_baseline.20260622.131551	hsa_full	inv_markup	baseline	kappa.0	0.0961	0.1934	-0.2090	0.4047	NaN	NaN	NaN	NaN	NaN
hsa_full_inv_markup_baseline.20260622.131551	hsa_full	inv_markup	baseline	delta	-0.0004	0.0012	-0.0028	0.0013	NaN	NaN	NaN	NaN	NaN
hsa_full_inv_markup_baseline.20260622.131551	hsa_full	inv_markup	baseline	theta.0	0.0669	0.0202	-0.2423	0.3712	NaN	NaN	NaN	NaN	NaN
hsa_full_inv_markup_baseline.20260622.131551	hsa_full	inv_markup	baseline	gamma	-0.0001	0.0010	-0.0015	0.0015	NaN	NaN	NaN	NaN	NaN
hsa_full_inv_markup_baseline.20260622.131551	hsa_full	inv_markup	baseline	kappa.t.mean	0.0049	0.1512	-0.2624	0.2615	NaN	NaN	NaN	NaN	NaN
hsa_full_inv_markup_baseline.20260622.131551	hsa_full	inv_markup	baseline	kappa.t.start	0.0066	0.1482	-0.2587	0.2588	NaN	NaN	NaN	NaN	NaN
hsa_full_inv_markup_baseline.20260622.131551	hsa_full	inv_markup	baseline	kappa.t.mid	-0.0028	0.1637	-0.2765	0.2761	NaN	NaN	NaN	NaN	NaN
hsa_full_inv_markup_baseline.20260622.131551	hsa_full	inv_markup	baseline	kappa.t.end	0.0163	0.1360	-0.2420	0.2447	NaN	NaN	NaN	NaN	NaN
hsa_full_inv_markup_baseline.20260622.131551	hsa_full	inv_markup	baseline	theta.t.mean	0.0460	0.0258	0.0113	0.1012	NaN	NaN	NaN	NaN	NaN
hsa_full_inv_markup_baseline.20260622.131551	hsa_full	inv_markup	baseline	theta.t.start	0.0465	0.0263	0.0089	0.0994	NaN	NaN	NaN	NaN	NaN
hsa_full_inv_markup_baseline.20260622.131551	hsa_full	inv_markup	baseline	theta.t.mid	0.0443	0.0289	-0.0115	0.1123	NaN	NaN	NaN	NaN	NaN
hsa_full_inv_markup_baseline.20260622.131551	hsa_full	inv_markup	baseline	theta.t.end	0.0487	0.0383	-0.0123	0.1066	NaN	NaN	NaN	NaN	NaN
hsa_full_inv_markup_baseline.20260622.131551	hsa_full	inv_markup	baseline	phi.1	0.6639	0.0583	0.5845	0.7842	NaN	NaN	NaN	NaN	NaN
hsa_full_inv_markup_baseline.20260622.131551	hsa_full	inv_markup	baseline	rho.1	1.1914	0.1130	1.0022	1.3879	NaN	NaN	NaN	NaN	NaN
hsa_full_inv_markup_baseline.20260622.131551	hsa_full	inv_markup	baseline	rho.2	-0.2695	0.1199	-0.4699	-0.0663	NaN	NaN	NaN	NaN	NaN
hsa_full_inv_markup_baseline.20260622.131551	hsa_full	inv_markup	baseline	n	-0.0899	0.0590	-0.1805	0.0371	NaN	NaN	NaN	NaN	NaN
hsa_full_inv_markup_baseline.20260622.131551	hsa_full	inv_markup	baseline	lambda.ez	-0.0122	0.1928	-0.3014	0.2785	NaN	NaN	NaN	NaN	NaN
hsa_full_inv_markup_baseline.20260622.131551	hsa_full	inv_markup	baseline	rho	-0.0095	0.1382	-0.2032	0.1983	NaN	NaN	NaN	NaN	NaN
hsa_full_inv_markup_baseline.20260622.131551	hsa_full	inv_markup	baseline	sigma.e	0.6852	0.0499	0.5984	0.7626	NaN	NaN	NaN	NaN	NaN
hsa_full_inv_markup_baseline.20260622.131551	hsa_full	inv_markup	baseline	sigma.zeta	0.4954	0.0328	0.4429	0.5677	NaN	NaN	NaN	NaN	NaN
hsa_full_inv_markup_baseline.20260622.131551	hsa_full	inv_markup	baseline	sigma.u	0.6964	0.0599	0.6001	0.8435	NaN	NaN	NaN	NaN	NaN
hsa_full_inv_markup_baseline.20260622.131551	hsa_full	inv_markup	baseline	sigma.eps	1.0612	0.0798	0.9286	1.1819	NaN	NaN	NaN	NaN	NaN
hsa_steady_inv_markup_baseline.20260622.131547	hsa_steady	inv_markup	baseline	alpha	0.6484	0.0637	0.5539	0.7759	NaN	NaN	NaN	NaN	NaN
hsa_steady_inv_markup_baseline.20260622.131547	hsa_steady	inv_markup	baseline	kappa.0	0.1008	0.1901	-0.1878	0.4623	NaN	NaN	NaN	NaN	NaN
hsa_steady_inv_markup_baseline.20260622.131547	hsa_steady	inv_markup	baseline	delta	-0.0006	0.0012	-0.0028	0.0021	NaN	NaN	NaN	NaN	NaN
hsa_steady_inv_markup_baseline.20260622.131547	hsa_steady	inv_markup	baseline	kappa.t.mean	-0.0290	0.1356	-0.2535	0.2345	NaN	NaN	NaN	NaN	NaN
hsa_steady_inv_markup_baseline.20260622.131547	hsa_steady	inv_markup	baseline	kappa.t.start	-0.0294	0.1379	-0.2596	0.2386	NaN	NaN	NaN	NaN	NaN
hsa_steady_inv_markup_baseline.20260622.131547	hsa_steady	inv_markup	baseline	kappa.t.mid	-0.0294	0.1362	-0.2593	0.2340	NaN	NaN	NaN	NaN	NaN
hsa_steady_inv_markup_baseline.20260622.131547	hsa_steady	inv_markup	baseline	kappa.t.end	-0.0286	0.1338	-0.2462	0.2266	NaN	NaN	NaN	NaN	NaN
hsa_steady_inv_markup_baseline.20260622.131547	hsa_steady	inv_markup	baseline	phi.1	0.6665	0.0599	0.5612	0.7780	NaN	NaN	NaN	NaN	NaN
hsa_steady_inv_markup_baseline.20260622.131547	hsa_steady	inv_markup	baseline	rho.1	1.8054	0.0517	1.7062	1.8953	NaN	NaN	NaN	NaN	NaN
hsa_steady_inv_markup_baseline.20260622.131547	hsa_steady	inv_markup	baseline	rho.2	-0.8090	0.0517	-0.8973	-0.7095	NaN	NaN	NaN	NaN	NaN
hsa_steady_inv_markup_baseline.20260622.131547	hsa_steady	inv_markup	baseline	n	-0.0538	0.0713	-0.1901	0.0504	NaN	NaN	NaN	NaN	NaN
hsa_steady_inv_markup_baseline.20260622.131547	hsa_steady	inv_markup	baseline	lambda.ez	0.0197	0.1538	-0.1838	0.3257	NaN	NaN	NaN	NaN	NaN
hsa_steady_inv_markup_baseline.20260622.131547	hsa_steady	inv_markup	baseline	rho	0.0172	0.1164	-0.1294	0.2546	NaN	NaN	NaN	NaN	NaN
hsa_steady_inv_markup_baseline.20260622.131547	hsa_steady	inv_markup	baseline	sigma.eta	0.6739	0.0385	0.6143	0.7469	NaN	NaN	NaN	NaN	NaN
hsa_steady_inv_markup_baseline.20260622.131547	hsa_steady	inv_markup	baseline	sigma.e	0.6785	0.0378	0.6219	0.7502	NaN	NaN	NaN	NaN	NaN
hsa_steady_inv_markup_baseline.20260622.131547	hsa_steady	inv_markup	baseline	sigma.zeta	0.5045	0.0317	0.4405	0.5570	NaN	NaN	NaN	NaN	NaN
hsa_steady_inv_markup_baseline.20260622.131547	hsa_steady	inv_markup	baseline	sigma.u	0.5105	0.0639	0.4296	0.6400	NaN	NaN	NaN	NaN	NaN
hsa_steady_inv_markup_baseline.20260622.131547	hsa_steady	inv_markup	baseline	sigma.eps	0.3901	0.0544	0.3255	0.5235	NaN	NaN	NaN	NaN	NaN
hsa_steady_inv_markup_baseline.20260622.131547	hsa_steady	inv_markup	baseline	sigma.N	0.2989	0.0219	0.2604	0.3341	NaN	NaN	NaN	NaN	NaN

data_spec	prior	parameter	mean	median	ci_2.5	ci_97.5	p_gt_0	p_lt_0	conclusion
inv_markup	baseline	alpha	0.6484	0.6341	0.5539	0.7759	1.0000	0.0000	NaN
inv_markup	baseline	kappa_0	0.1008	0.0783	-0.1878	0.4623	0.6750	0.3250	NaN
inv_markup	baseline	delta	-0.0006	-0.0007	-0.0028	0.0021	0.3000	0.7000	NaN
inv_markup	baseline	rho_1	1.8054	1.7946	1.7062	1.8953	1.0000	0.0000	NaN
inv_markup	baseline	rho_2	-0.8090	-0.7989	-0.8973	-0.7095	0.0000	1.0000	NaN
inv_markup	baseline	sigma_e	0.6785	0.6781	0.6219	0.7502	1.0000	0.0000	NaN
inv_markup	baseline	sigma_zeta	0.5045	0.4964	0.4495	0.5570	1.0000	0.0000	NaN
inv_markup	baseline	sigma_u	0.5105	0.4864	0.4296	0.6400	1.0000	0.0000	NaN
inv_markup	baseline	sigma_eps	0.3901	0.3884	0.3255	0.5235	1.0000	0.0000	NaN
inv_markup	baseline	sigma_N	0.2989	0.2976	0.2604	0.3341	1.0000	0.0000	NaN
inv_markup	weak	alpha	0.6458	0.6486	0.5432	0.7534	1.0000	0.0000	NaN
inv_markup	weak	kappa_0	0.2063	0.1583	-0.7736	1.3392	0.6250	0.3750	NaN
inv_markup	weak	delta	-0.0009	-0.0009	-0.0066	0.0041	0.3750	0.6250	NaN
inv_markup	weak	rho_1	1.9069	1.9105	1.8328	1.9713	1.0000	0.0000	NaN
inv_markup	weak	rho_2	-0.9096	-0.9116	-0.9730	-0.8367	0.0000	1.0000	NaN
inv_markup	weak	sigma_e	0.6771	0.6717	0.6029	0.7878	1.0000	0.0000	NaN
inv_markup	weak	sigma_zeta	0.5022	0.4988	0.4480	0.5509	1.0000	0.0000	NaN
inv_markup	weak	sigma_u	0.4183	0.4162	0.3521	0.4827	1.0000	0.0000	NaN
inv_markup	weak	sigma_eps	0.3040	0.3035	0.2409	0.3625	1.0000	0.0000	NaN
inv_markup	weak	sigma_N	0.2268	0.2215	0.1957	0.2747	1.0000	0.0000	NaN
inv_markup	tight	alpha	0.6124	0.6078	0.5147	0.7208	1.0000	0.0000	NaN
inv_markup	tight	kappa_0	0.0996	0.1066	-0.1344	0.2868	0.8750	0.1250	NaN
inv_markup	tight	delta	-0.0008	-0.0007	-0.0023	0.0005	0.1750	0.8250	NaN
inv_markup	tight	rho_1	0.5521	0.5496	0.3924	0.6863	1.0000	0.0000	NaN
inv_markup	tight	rho_2	-0.3597	-0.3573	-0.5528	-0.1244	0.0000	1.0000	NaN
inv_markup	tight	sigma_e	0.6758	0.6708	0.6194	0.7495	1.0000	0.0000	NaN
inv_markup	tight	sigma_zeta	0.4880	0.4869	0.4428	0.5439	1.0000	0.0000	NaN
inv_markup	tight	sigma_u	0.5543	0.5562	0.4690	0.6687	1.0000	0.0000	NaN
inv_markup	tight	sigma_eps	1.2632	1.2487	1.1261	1.4178	1.0000	0.0000	NaN
inv_markup	tight	sigma_N	0.5328	0.5308	0.4439	0.6412	1.0000	0.0000	NaN
inv_markup	all	delta_robustness	NaN	NaN	NaN	NaN	NaN	NaN	robust

note

Run scripts/05_period_robustness.py first.

11 Posterior Predictive Checks and Figures

Posterior predictive checks are generated after model runs.

12 Notes

WAIC and LOO are not used as primary criteria because the models contain latent states. Chib marginal likelihoods are reported only when the legacy ordinate calculation can be matched to the sampler units and supplied data.